Lecture 09

Carbon Tariffs

Dana Andersen
ECON 467

Roadmap

1. What are carbon tariffs?
2. Basic trade theory: supply, demand, and tariffs
3. Adding pollution externalities to trade
4. Unilateral carbon taxes and leakage
5. Carbon tariffs to prevent leakage
6. Carbon tariffs alone
7. World carbon taxes
8. Different pollution intensities across countries

What are carbon tariffs?

Carbon tariffs

Carbon tariffs are tariffs imposed on imports based on their **carbon content**

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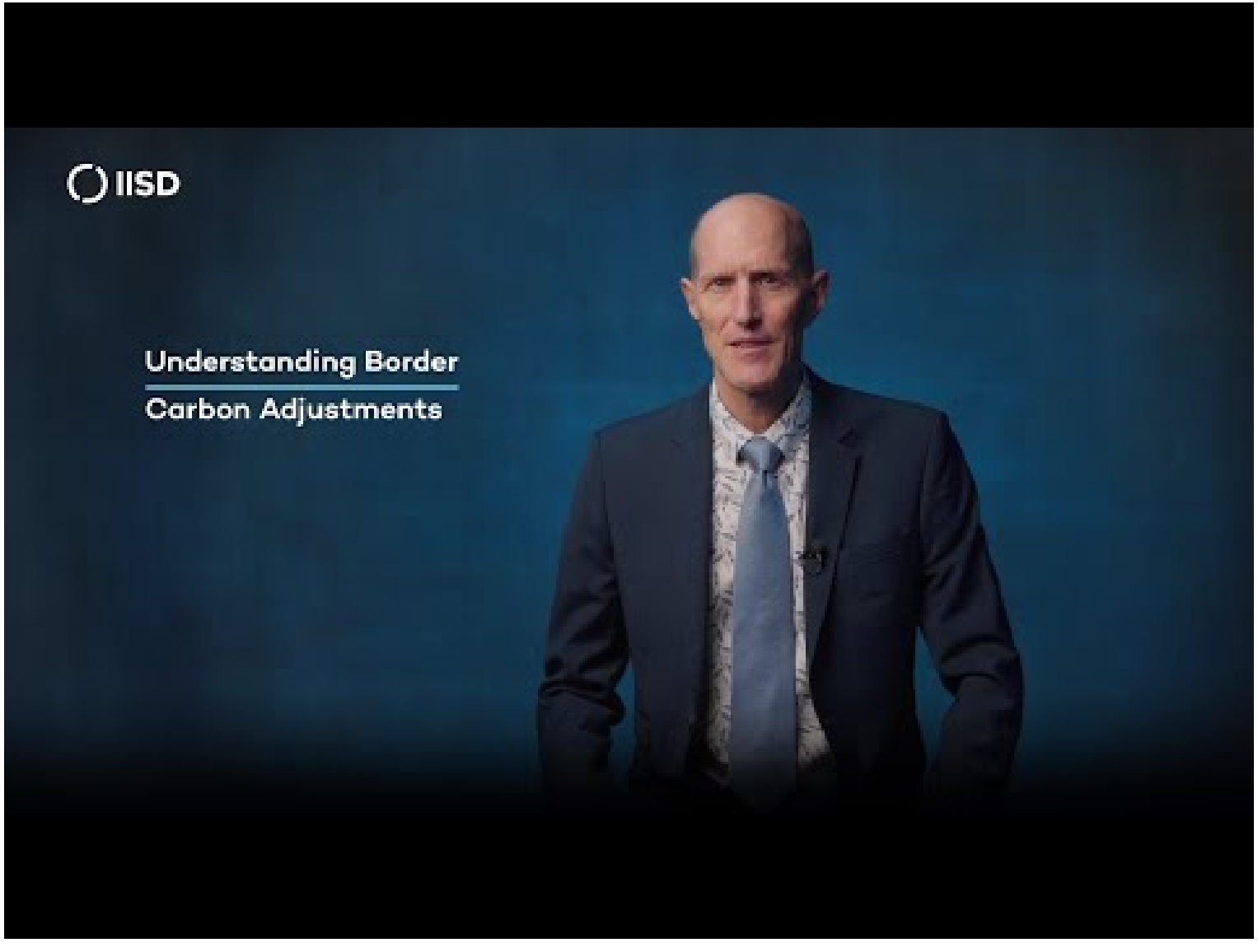
Why would we want carbon tariffs?

Understanding border carbon adjustments



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Carbon tariffs aim to "level the playing field" and reduce leakage

Carbon tariffs in practice

EU Carbon Border Adjustment Mechanism (CBAM):

- Phasing in 2023-2026, full implementation 2026
- Covers cement, iron/steel, aluminum, fertilizers, electricity, hydrogen
- Importers must purchase certificates matching EU ETS price

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UK CBAM: announced for 2027

US proposals: various bills proposed but none passed yet

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1. How carbon tariffs work in theory
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The key insight: carbon tariffs are equivalent to a **consumption tax** plus a **production subsidy**

Basic trade theory

A simple trade model

Consider a small open economy (country i) that:

- Takes world prices as given
- Can import or export a homogeneous good

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Key curves:

- S: domestic supply
- D: domestic demand
- p_w: world price

Free trade equilibrium

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## [Figure 1: Standard tariff - S, D curves, deadweight triangles a, b]
```

With free trade at price p_w:

- Domestic production: Q_1
- Domestic consumption: Q_4
- Imports: $Q_4 - Q_1$

Adding a tariff

A tariff t raises the domestic price to $p_w + t$

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- Domestic production rises: $Q_1 \to Q_2$
- Domestic consumption falls: $Q_4 \to Q_3$
- Imports fall: $(Q_4 - Q_1) \to (Q_3 - Q_2)$

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Deadweight loss: triangles $a + b$

- a: production distortion (inefficient domestic production)
- b: consumption distortion (foregone consumer surplus)

Tariff revenue

The tariff generates revenue: $t \times (Q_3 - Q_2)$

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Standard result: **tariffs reduce welfare** in a small open economy

Adding pollution

Production creates pollution

Now suppose production of this good creates pollution

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MSC lies above the private supply curve by the external cost $e \times d$

Free trade with pollution

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## [Figure 2a: Country i with MSC curve above S]
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Country i's MSC curve lies above its supply curve

The gap is the external cost: $e \times d$

Global perspective

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## [Figure 2b: World S, MSC, D curves]
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From a global perspective:

- Free trade output: where $D = S$ (too much!)
- Efficient output: where $D = MSC$ (less production)

The externality problem

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But what if only **one country** acts?

Unilateral carbon tax

Country i imposes a carbon tax

Suppose country i imposes a carbon tax $\tau = e \times d$ on domestic production

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But the **world price doesn't change** (small open economy)

Effects of unilateral carbon tax

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## [Figure 3: Unilateral carbon tax with leakage]
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Unilateral carbon tax effects

Country i's carbon tax:

- Reduces domestic production
- Does not change domestic consumption (price unchanged)
- **Increases imports**

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This is **carbon leakage**: emissions shift abroad rather than being reduced

Leakage undermines the tax

With identical carbon intensities:

- Every unit of reduced domestic production = 1 unit more foreign production
- **Net global emissions unchanged!**

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The carbon tax:

- Generates tax revenue R_t (rectangle)
- Creates deadweight loss d (triangle) from production distortion
- Achieves **zero environmental benefit**

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Country i bears costs but gets no environmental gain

Carbon tax + carbon tariff

Adding a carbon tariff

Now suppose country i adds a carbon tariff equal to the carbon tax:

$t = \tau = e \times d$

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Domestic price rises to: $p_w + t$

Carbon tax + tariff effects

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## [Figure 4: Carbon tax + tariff effects]
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How the tariff helps

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Net effect:

- Production distortion from tax alone: $-d$
- Consumption distortion from tariff: $-b$
- **Environmental benefit:** reduced global emissions

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Net effect:

- Production distortion from tax alone: $-d$
- Consumption distortion from tariff: $-b$
- **Environmental benefit:** reduced global emissions

If environmental benefit $> d + b$: net welfare gain!

Numerical check (practice-style)

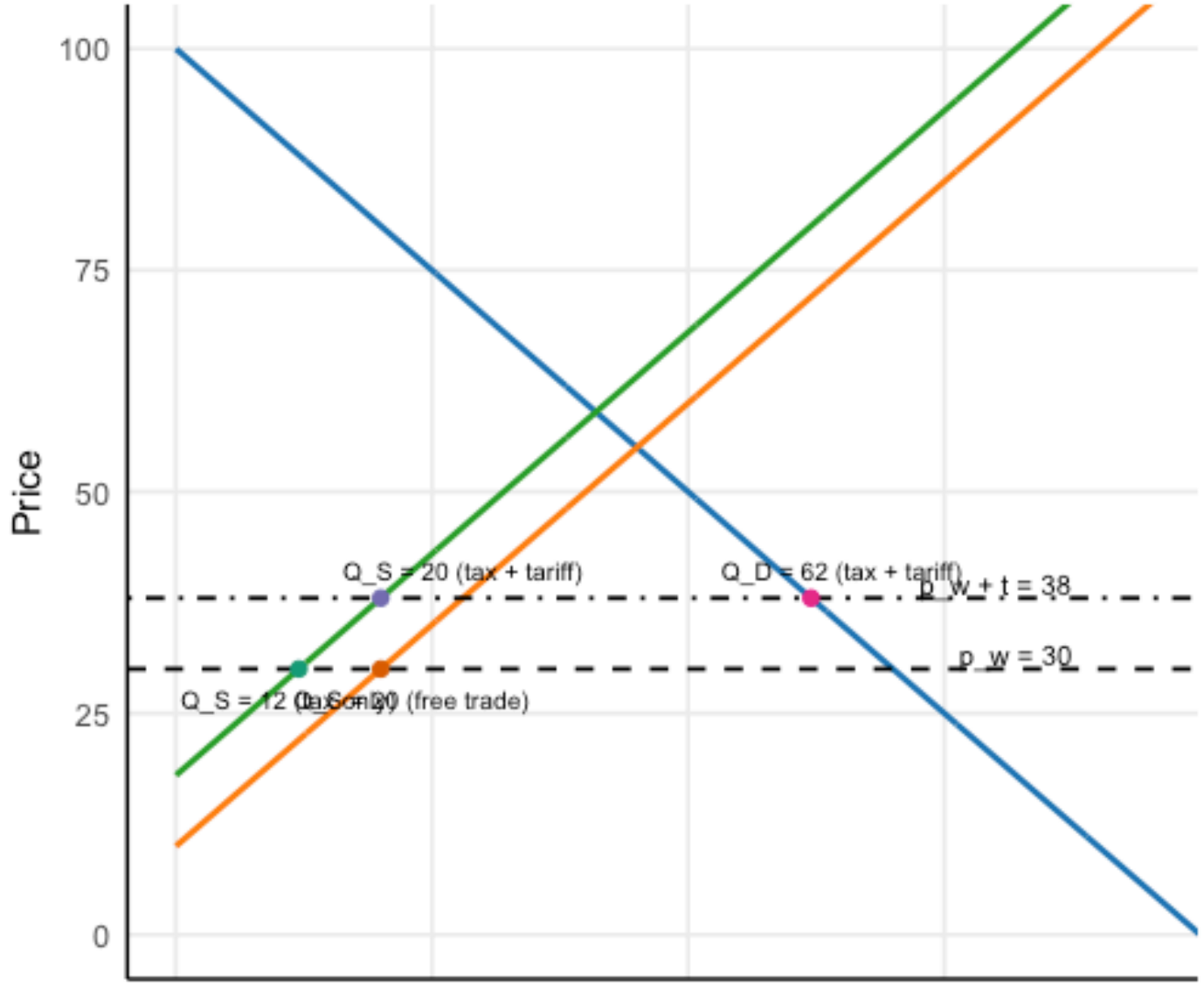
Use the same small-open-economy setup as in the practice set:

- $Q_D = 100 - P$
- $Q_S = P - 10$
- $p_w = 30$
- Identical carbon intensities: $e_H = e_F = 1$
- Marginal damage per ton: $d = 8$
- Domestic carbon tax: $\tau = e_H d = 8$
- Border tariff with tax: $t = \tau = 8$

Compute under:

1. Free trade

Numerical check: graph



Numerical check: answers

1. **Free trade at `(P=30)`**

- $Q_S = 20,\ Q_D = 70,\ M = 50$
- Emissions: $E_H = 20,\ E_F = 50,\ E_{global} = 70$

1. **Unilateral domestic tax only (`(\tau=8)`)**

- Domestic supply shifts to $Q_S = P - 18$
- At $P = 30$: $Q_S = 12,\ Q_D = 70,\ M = 58$
- Emissions: $E_H = 12,\ E_F = 58,\ E_{global} = 70$ (pure leakage)

1. **Domestic tax + tariff (`(\tau=t=8)`)**

- Domestic price rises to $P = p_w + t = 38$

The key insight

With identical carbon intensities everywhere:

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Carbon tax + carbon tariff = **Consumption tax**

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Why?

- Tax on domestic production raises domestic costs
- Tariff on imports raises import costs
- Both domestic and imported goods face the same carbon price
- Equivalent to taxing consumption of the carbon-intensive good

Carbon tariff alone

What if we only use a tariff?

Some proposals call for carbon tariffs **without** a domestic carbon tax

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What does this accomplish?

Carbon tariff alone

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## [Figure 5: Carbon tariff alone = consumption tax + production subsidy]
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Domestic emissions may **rise**!

Is a tariff alone good policy?

A carbon tariff alone:

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- But **subsidizes** domestic carbon-intensive production (bad)

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May violate WTO rules (discriminates against foreign producers)

World carbon tax

What if everyone taxes carbon?

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No leakage: all production faces the same carbon price

World carbon tax

```
## [Figure 6: World carbon tax - efficient outcome]
```

Efficient outcome

With a world carbon tax at $\tau = e \times d$:

- Global production equals efficient level
- Price reflects full social cost
- **No need for carbon tariffs**

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Carbon tariffs are a **second-best** policy for when global coordination fails

Insufficient world carbon tax

What if the world carbon tax is **too low**?

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Suppose world tax = $\tau' < e \times d$

Insufficient world carbon tax

What if the world carbon tax is **too low**?

Suppose world tax = $\tau' < e \times d$

Global emissions still too high, but less so than with no tax

Too-small world tax

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## [Figure 7: Too-small world carbon tax]
```

Can country i improve things with a tariff?

Tariff on top of insufficient world tax

```
## [Figure 8: Overshooting with tariff on insufficient world tax]
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Overshooting problem

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A tariff can't fix insufficient global carbon pricing

Different carbon intensities

What if countries differ?

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- Different production technologies
- Different regulatory environments

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How should tariffs account for this?

Cleaner domestic production

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Now the **efficient** policy differs for domestic vs. foreign goods

Tariff based on domestic carbon intensity

```
## [Figure 9: Tariff based on domestic carbon intensity]
```

If tariff = domestic carbon tax = $e_i \times d$:

- Correctly prices domestic emissions
- **Under-prices** foreign emissions (since $e_j > e_i$)

Tariff based on foreign carbon intensity

```
## [Figure 10: Tariff based on foreign carbon intensity]
```

If tariff = $e_j \times d$ (foreign intensity):

- Correctly prices foreign emissions
- But tariff $>$ domestic tax: **looks protectionist**

The measurement problem

Setting tariffs based on actual foreign carbon intensity requires:

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2. Verifying foreign carbon prices
3. Adjusting for different products/producers

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1. Measuring emissions in foreign production
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This is **extremely difficult** in practice

EU CBAM uses default values, allows importers to prove lower emissions

Complications

Real-world complications

Several factors complicate carbon tariff design:

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3. **Export rebates:** should exports get carbon tax rebates?

Real-world complications

Several factors complicate carbon tariff design:

1. **Resource shuffling:** clean production exported, dirty production consumed domestically
2. **Non-price policies:** regulations vs. carbon prices
3. **Export rebates:** should exports get carbon tax rebates?
4. **Retaliation:** trading partners may impose counter-tariffs

Resource shuffling

Suppose country j has both clean and dirty production

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With a carbon tariff based on average intensity:

- Export the clean stuff to country i
- Consume the dirty stuff domestically

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This is **another form of leakage**

Non-price policies

Many countries use regulations rather than carbon prices:

- Fuel economy standards
- Renewable portfolio standards
- Building codes

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How do you calculate the "equivalent carbon price"?

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How do you calculate the "equivalent carbon price"?

Very difficult: may require complex modeling

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Should country i rebate carbon taxes on exports?

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Arguments against:

- Reduces domestic incentive to decarbonize
- May look like an export subsidy (WTO issues)

WTO compatibility

Carbon tariffs face WTO scrutiny:

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EU CBAM designed carefully to fit within WTO rules

Key takeaways

Summary

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2. **Carbon tax + tariff** is equivalent to a consumption tax

   - Reduces consumption of carbon-intensive goods
   - Can improve welfare if environmental benefit exceeds distortion costs

3. **Carbon tariff alone** = consumption tax + production subsidy

   - May increase domestic emissions
   - Less effective than tax + tariff

Summary (continued)

1. **World carbon tax** is first-best
   - Carbon tariffs are second-best when global coordination fails

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1. **World carbon tax** is first-best
   - Carbon tariffs are second-best when global coordination fails
2. **Different carbon intensities** create measurement challenges
   - Setting tariff = foreign intensity is efficient but looks protectionist
   - Setting tariff = domestic intensity under-prices foreign emissions

Summary (continued)

1. **World carbon tax** is first-best
   - Carbon tariffs are second-best when global coordination fails
2. **Different carbon intensities** create measurement challenges
   - Setting tariff = foreign intensity is efficient but looks protectionist
   - Setting tariff = domestic intensity under-prices foreign emissions
3. **Many complications** in practice
   - Resource shuffling, non-price policies, export rebates, WTO rules

The bottom line

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- Set based on actual carbon content
- Part of broader international coordination

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Alone, they are an imperfect tool that may create new distortions

References

Brunel, C. and A. Levinson (2024). "Carbon Tariffs 101." NBER Working Paper 33024.

EU Carbon Border Adjustment Mechanism: https://taxation-customs.ec.europa.eu/carbon-border-adjustment-mechanism_en